The probabilities are given by

$$f(0) = P(X=0) = \frac{1}{4},$$

$$f(1) = P(X=1) = \frac{1}{2}$$

and $$f(2) = P(X=2) = \frac{1}{4}$$

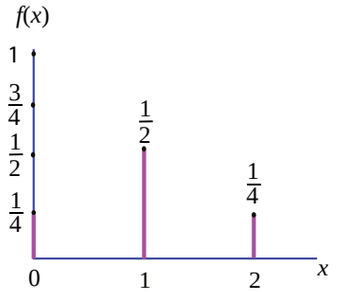
The function $f(x)$ satisfies the conditions

(i) $f(x) \geq 0$, for $x = 0, 1, 2$

(ii) $$\sum_x f(x) = \sum_{x=0}^{x=2} f(x) = f(0) + f(1) + f(2)$$

$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$$

Therefore $f(x)$ is a probability mass function.



Probability mass function of $f(x)$

Fig. 11.4

The probability mass function is given by

x	0	1	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

(or)

$$f(x) = \begin{cases} \frac{1}{4} & \text{for } x = 0 \\ \frac{1}{2} & \text{for } x = 1 \\ \frac{1}{4} & \text{for } x = 2 \end{cases}$$

Example 11.6

A pair of fair dice is rolled once. Find the probability mass function to get the number of fours.

Solution

Let X be a random variable whose values x are the number of fours.

The sample space S is given in the table.

$$S = \begin{Bmatrix} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{Bmatrix}$$

It can also be written as

$S = \{(i, j)\}$, where $i = 1, 2, 3 \ldots, 6$ and $j = 1, 2, 3, \ldots 6$

Therefore X takes on the values of 0, 1, and 2.

We observe that

(i) $X = 0$, if (i, j) for $i \neq 4$, $j \neq 4$,

(ii) $X = 1$, if $(1,4), (2,4), (3,4), (5,4), (6,4), (4,1), (4,2), (4,3), (4,5), (4,6)$

(iii) $X = 2$, if $(4,4)$,

Therefore,

Values of the Random Variable X	0	1	2	Total
Number of elements in inverse images	25	10	1	36

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